PRIVILEGED & CONFIDENTIAL — ATTORNEY WORK PRODUCT — PREPARED FOR COUNSEL IN ANTICIPATION OF A PROVISIONAL PATENT APPLICATION

IP Disclosure Package

The Investment Map — parcel-level corridor investment decision system

INVENTOR	Christa D. Stoneham (Connecting Dots and Strategies)
PREPARED	August 3, 2026
SUBJECT CODEBASE	Repository `christastoneham-source/Christa-Claude-`, files `McNichols_Kresge_DEMO.html` (v1), `McNichols_Kresge_DEMOv2.html` (v2, primary), `McNichols_Kresge_DEMO_OFFLINE.html`
STANDARD APPLIED	Accuracy over polish. Formulas quoted verbatim with file and line citations. Anything unverifiable is marked **UNVERIFIED** with what would confirm it. Nothing sanitized.
HEADLINE FLAGS	(1) Earliest known disclosure AND same-day sale: **May 4, 2026** (email of record, §2). (2) The invention predates the repository ("ai-planner" predecessor). (3) Built vs. intended embodiments are explicitly separated throughout.

CONTENTS

1. Invention Timeline

Prepared 2026-08-03 for provisional patent disclosure. Attorney work product. All git output below is verbatim from the repository `christastoneham-source/Christa-Claude-`, branch `claude/wizardly-meitner-eigjm9`. Where a date is inferred rather than recorded in git, it is marked INFERRED.

⚠️ FLAG: EARLIEST KNOWN OUTSIDE DISCLOSURE — DOCUMENTED BY EMAIL, WITH A SAME-DAY SALE

Earliest known external disclosure: Monday, 2026-05-04, 2:15 PM — the inventor emailed the working tool (HTML file + user guide, 2 attachments) to Tosha M. Tabron, Director of Originations, Social Investment Practice, The Kresge Foundation. The full email thread is transcribed in `EVIDENCE-2026-05-04-EMAIL.md` in this folder. The email contains no confidentiality language, and the recipient replied the same day that she would show it to her team the next day (2026-05-05).

⚠️ **SAME-DAY COMMERCIAL SALE.** The delivery email priced the phase at $1,500; the recipient accepted at 9:48 PM the same day, committing $2,500 "for this first phase." **Counsel should analyze 2026-05-04 as BOTH a public disclosure AND a potential on-sale bar event (35 U.S.C. § 102(a)(1))** — the on-sale analysis can apply even where a disclosure is arguably non-public.

This event predates the repository (first repo commit 2026-06-15; tool's first commit here 2026-06-29, "forked from ai-planner"). The May 4 artifact was an ai-planner-era build. Because the tool is a single HTML file, sending it conveyed the complete working system, its formulas, and its embedded data — a stronger disclosure posture than a supervised demo.

Consequences for counsel: - **Conservative US one-year bar date: on or about 2027-05-04** (grace period from the May 4 disclosure/sale), not 2027-07-03 as previously inferred from git alone. - Foreign absolute-novelty analysis should assume a 2026-05-04 disclosure; no confidentiality restriction is visible in the thread. - **The content of the May 4 attachment defines WHAT was disclosed/sold that day.** Per the email's own description, the May 4 build included: the four pillars, click-to-assign with immediate projected impact, the square-footage-driven impact calculator, per-parcel Year-1/Year-2/long-term outcome narratives, and Google Maps links — i.e., the CORE coefficient engine. It explicitly did NOT yet have accurate spatial/geographic layout, and GIS integration was named as a future phase. Not mentioned scenario save/compare, dashboards, value/tax projection. Features added in this repo (editable pillars and cost overrides 2026-06-29; scenarios/share files, real parcel geometry, building footprints, value/tax card, offline build 2026-07-03) have their own later dates unless present in the May 4 attachment.

ACTION ITEMS: see the preservation checklist in `EVIDENCE-2026-05-04-EMAIL.md` (original .eml with headers from both email accounts, the two exact attachments, the $2,500 payment record, the rest of the 9-message thread, and the ai-planner repository).

UNVERIFIED: the exact contents of the May 4 attachments; whether/when the $2,500 was paid; the remainder of the email thread (the provided message was "5 of 9" in the thread).

Secondary evidence previously inferred from git (superseded as "earliest," retained as corroboration of later events)

A Kresge Foundation demo on or about 2026-07-03 remains the earliest disclosure event evidenced INSIDE this repository:

1. The tool itself is addressed to an outside party from its first commit. The guided tour text in the code reads "A quick tour of this corridor decision demo, built for The Kresge Foundation" (`McNichols_Kresge_DEMOv2.html:802`), and the PDF export header reads "Prepared for The Kresge Foundation" (`McNichols_Kresge_DEMOv2.html:792`). The same Kresge-facing language exists in the v1 file committed 2026-06-29 (commit 913a55a).
2. `McNichols_DEMO_TODO.md:2` (committed 2026-07-03 23:47 UTC) says: "Saved 2026-07-03. Items deliberately NOT built before the Kresge demo" — written in anticipation of a demo.
3. An offline backup build was created 2026-07-03 23:28 UTC (commit 5f93f95, "Add offline backup build with Leaflet, Chart.js, and html2canvas embedded") — the kind of artifact prepared immediately before presenting in a room with uncertain internet.
4. The next commit after the 2026-07-03 cluster (24d2a64, 2026-07-06) defers new work "until contract," consistent with the demo having occurred between 2026-07-03 and 2026-07-06.

The Kresge demo date itself is NOT recorded in git. UNVERIFIED: the meeting date, attendees, and confidentiality status. Given the inventor's 2026-05-04 attestation above, the operative earliest-disclosure date for bar-date purposes is 2026-05-04; the ~2026-07-03 Kresge demo is a second, later disclosure event covering the more evolved v2 feature set.

Secondary disclosure vectors visible in the repo, each UNVERIFIED as to whether it was exercised: - The scenario Share/Load feature is designed to send files to other people ("sends it by email or Teams", `McNichols_Kresge_DEMOv2.html:609`). - The one-pager prompt (`cds-investment-map-onepager-prompt.md`, 2026-07-13) references a marketing one-pager PDF that predates it and describes the offering publicly-facing language; whether that PDF was distributed is not in the repo.

⚠️ FLAG: THE INVENTION PREDATES THIS REPOSITORY

The first commit of the tool (913a55a, 2026-06-29) says it was **"forked from ai-planner."** The predecessor "ai-planner" codebase is NOT in this repository. Conception and some reduction to practice occurred before 2026-06-29 in that predecessor. UNVERIFIED: the ai-planner repository's location, dates, and contents. The inventor should preserve and produce that history to counsel; it may move conception dates earlier (helpful) and may contain its own disclosure events (must be checked).

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1. Full commit history, oldest first

Command: `git log --reverse --date=iso --pretty=format:"%ad %h %s"`

Verbatim output:

```
2026-06-15 02:16:30 +0000 4a21d86 Add 7811 Harrisburg RFP review workspace: workflow, templates, and reference materials
2026-06-15 02:30:48 +0000 eff0eba Add Lovable build prompt for the 7811 Harrisburg RFP reviewer platform
2026-06-29 06:00:50 +0000 913a55a Add McNichols Kresge corridor demo (forked from ai-planner, locked to W. McNichols, CDS rebrand)
2026-06-29 06:16:39 +0000 52c4e9c Add 24 not-in-scope context parcels to corridor schematic and map; keep them excluded from all tallies/export
2026-06-29 06:32:21 +0000 336bbf3 Demo: editable pillars + per-parcel cost, existing assessor use, property value/tax-base impact on dashboard
2026-06-29 06:49:12 +0000 edeee1a Add Marygrove Conservancy as green not-in-scope feature; add global cost-adjustment sensitivity; existing lan
2026-07-03 18:57:38 +0000 354f07d Add v2 build: scenario export/import, provenance stamp + versioning, exempt-parcel handling, Kresge navy/gold
2026-07-03 19:16:44 +0000 103ebc9 Scenarios: plain-language share/load + visual corridor strips + tax compare row; dashboard tax card; live bui
2026-07-03 19:30:33 +0000 3a2eb87 Add building floor area + year built of record to parcel detail; harden footprint layer with multiple sources
2026-07-03 21:46:05 +0000 c88392d Fix 7303 condo parcel: merge both unit geometries + combined assessed value; merge Detroit + OSM footprint so
2026-07-03 23:28:32 +0000 5f93f95 Add offline backup build with Leaflet, Chart.js, and html2canvas embedded
2026-07-03 23:41:15 +0000 2a204e3 Cache building footprints in browser after first successful load; footprints then persist offline and across
2026-07-03 23:47:16 +0000 2cbbfdf Add parked to-do list for the McNichols demo (post-Kresge work items)
2026-07-03 23:57:44 +0000 be44a89 Show pillar assignment symbols on Map View parcels
2026-07-06 01:25:58 +0000 24d2a64 Add approved environmental screening approach (EGLE-first, three rings) to demo TODO; build deferred until co
2026-07-13 20:22:22 +0000 8ac46d8 Add architectural visualization alignment assessment (data vs rendering, three-views frame) to TODO
2026-07-13 23:37:44 +0000 353da8c Add Investment Map one-pager design prompt with brand alignment notes
```

2. First commit

- **Repository first commit:** 2026-06-15 02:00:52 +0000, f8fe2a1, "Add Harris County Heirs' Property Navigator design package." This commit is a DIFFERENT project (see Scope note below), not the Investment Map.
- **First commit of the invention:** 2026-06-29 06:00:50 +0000, 913a55a, "Add McNichols Kresge corridor demo (forked from ai-planner, locked to W. McNichols, CDS rebrand)" — adds `McNichols_Kresge_DEMO.html` complete with the pillar-assignment engine, cost/outcome projection, dashboard, scenarios, map, AMI reference, and readiness checklist already present. This confirms substantial pre-repository development (see flag above).

3. Capability first-appearance dates (traced by commit)

All dates documented in git unless marked otherwise.

(UTC)			
≤2026-06-29	Corridor schematic (frontage-scaled parcel band), 4 investment pillars, per-parcel pillar assignment, sqft-driven cost and outcome projection, live tally/narrative, dashboard charts, cost-per-outcome table, property value/tax projection inputs (valueFactor, taxRate), AMI reference table, scenario save/load/compare, Leaflet map with real parcel polygons + satellite, zoning color mode, PNG/PDF export, localStorage persistence, guided tour, readiness checklist (later removed)	913a55a	file present in `git show 913a55a:McNichols_Kresge_DEMO.html` ; features predate repo (ai-planner fork)
2026-06-29	Not-in-scope context parcels excluded from all tallies	52c4e9c	commit message + `buildContext()`
2026-06-29	Editable pillars (user-defined pillar costs/metrics), per-parcel cost override, existing assessor use display, dashboard property value + tax base impact section; readiness checklist REMOVED	336bbf3	commit message; `parcelCost` gains `costOverride`
2026-06-29	Global cost-adjustment sensitivity factor (`assume.costFactor`)	edeee1a	commit message
2026-07-03	v2 file: scenario export/import (share file), provenance stamp + `TOOL_VERSION` , exempt-parcel handling	354f07d	adds `McNichols_Kresge_DEMOv2.html`
2026-07-03	Visual corridor mini-strips in scenario compare, scenario tax compare row, live building-footprint overlay (Detroit open data)	103ebc9	commit message
2026-07-03	Building floor area + year built of record in parcel detail; multi-source footprint fallback	3a2eb87	commit message
2026-07-03	Multi-source footprint merge with centroid dedup (Detroit + OSM); condo multi-unit parcel merge	c88392d	commit message
2026-07-03	Offline backup build (libraries embedded)	5f93f95	adds `McNichols_Kresge_DEMO_OFFLINE.html`
2026-07-03	Browser cache of building footprints (offline persistence of fetched GIS data)	2a204e3	commit message; `BLD_CACHE_KEY`
2026-07-03	Pillar assignment symbols on map parcels	be44a89	commit message
2026-07-06	Environmental screening design (three-ring architecture; documented, NOT built)	24d2a64	`McNichols_DEMO_TODO.md`
2026-07-13	Architectural visualization design (three-altitude frame; documented, NOT built)	8ac46d8	`McNichols_DEMO_TODO.md`

4. References to demos, clients, deploys, share links, presentations

- Commit 913a55a (2026-06-29): "McNichols **Kresge** corridor demo" — client-named demo.
- Commit 336bbf3 (2026-06-29): "**Demo**: editable pillars..."
- Commit 354f07d (2026-07-03): "**Kresge** navy/gold chrome" — client-branded UI.
- Commit 103ebc9 (2026-07-03): "plain-language **share**/load".
- Commit 5f93f95 (2026-07-03): "offline **backup** build" (presentation contingency).
- Commit 2cbbfdf (2026-07-03): "parked to-do list for the McNichols **demo** (post-**Kresge** work items)".
- Commit 24d2a64 (2026-07-06): "build deferred until **contract**".
- In-code: "Prepared for The Kresge Foundation" (`McNichols_Kresge_DEMOv2.html:168, 792`), tour step "built for The Kresge Foundation" (`:802`).
- Branch names: `claude/wizardly-meitner-eigjm9` (this work), `claude/epic-bardeen-er1ai6` (a separate Heirs' Property web build for Houston Land Bank, 2026-06-15 — different project). No tags exist.

5. Deployment configuration, hosting config, public URLs

- **None found.** No Netlify/Vercel/GitHub Pages/hosting config files exist in the repository. `McNichols_DEMO_TODO.md:110-111` records a hosted link (Netlify) as a PLANNED item "when Kresge says yes" — i.e., not deployed as of 2026-07-13.
- The tool loads libraries and map data from public CDNs/APIs at runtime (unpkg, cdnjs, Google Fonts, OSM/Esri tiles, Detroit ArcGIS, Overpass); these are inbound dependencies, not deployments of the tool.

6. Exported files, screenshots, PDFs, build artifacts in the repo

- `McNichols_Kresge_DEMO_OFFLINE.html` — build artifact (libraries inlined), added 2026-07-03 23:28 UTC (5f93f95). This is the only build artifact.
- No screenshots, no PDFs, no exported scenario JSON files are committed.

`:770`, print-to-PDF report at `:777–798`, scenario share file `W-McNichols-scenario-share.json` at `:558`); none are stored in the repo.
- UNVERIFIED: a marketing one-pager PDF exists outside the repo (referenced in conversation and by `cds-investment-map-onepager-prompt.md`) — obtain and date it separately.

7. Chronology table

DATE	MILESTONE	EVIDENCE	DOCUMENTED OR INFERRED
Before 2026-05-04	Conception + first reduction to practice in predecessor "ai-planner" (a working iteration existed by May 4)	Fork note in 913a55a commit message; inventor attestation below	INFERRED (predecessor repo UNVERIFIED, not in this repo)
2026-05-04 (Mon)	**Tool delivered by email to The Kresge Foundation (T. Tabron) with user guide; priced at $1,500; accepted same day at $2,500 — earliest known disclosure AND sale**	Email thread transcribed in `EVIDENCE-2026-05-04-EMAIL.md`; not in git	DOCUMENTED (email of record held by inventor; original .eml + attachments to be preserved)
2026-06-15	Repository created; two unrelated projects committed	f8fe2a1, 4a21d86	Documented
2026-06-29	Investment Map v1 committed: full decision engine present (pillars, projections, dashboard, scenarios, map, AMI)	913a55a	Documented
2026-06-29	Same-day iteration: context parcels, editable pillars, per-parcel cost override, value/tax projection UI, sensitivity factor; readiness checklist removed	52c4e9c, 336bbf3, edeee1a	Documented
2026-07-03	v2: share files, provenance stamp, exempt handling, corridor-picture compare, live GIS footprints, floor area of record, offline build, footprint caching	354f07d…be44a89 (7 commits)	Documented
~2026-07-03/07	Kresge Foundation demo (second known external showing; v2 feature set)	TODO header; offline backup timing; "until contract" on 07-06	INFERRED — confirm actual date
2026-07-06	Environmental screening architecture documented (intended embodiment)	24d2a64	Documented
2026-07-13	Architectural visualization architecture documented (intended embodiment); marketing one-pager prompt	8ac46d8, 353da8c	Documented

Scope note

`README.md`, `docs/` (Harris County Heirs' Property Navigator) and `hlb-7811-harrisburg-rfp/` (RFP review workspace), plus branch `claude/epic-bardeen-er1ai6`, are separate projects that happen to live in the same repository. They are OUT OF SCOPE for this disclosure and are not part of the Investment Map invention. The repo's `README.md` describes the Heirs' Property project, not the Investment Map.

2. Evidence: May 4, 2026 Email

Privileged and confidential — attorney work product. Transcribed 2026-08-03 from the inventor's Gmail thread as provided by the inventor. Gmail interface text has been trimmed; the message bodies are reproduced verbatim. The inventor notes the thread was later transferred from christastoneham@gmail.com to her other business email account — BOTH copies should be preserved.

Message 1 — delivery and offer

- **From:** christa stoneham christastoneham@gmail.com
- **To:** Tosha M. Tabron TMTabron@kresge.org, Director of Originations, Social Investment Practice, The Kresge Foundation
- **Date:** Mon, May 4 [2026], 2:15 PM
- **Subject:** McNichols Corridor Decision Tool: Delivery and Investment Summary
- **Attachments:** 2 (the tool HTML file and "a brief user guide") — **the attached files themselves are NOT in this repository; preserving the exact attachments is a priority action item.**

> *Hi Tosha,*
>
> *I am pleased to deliver the McNichols Corridor Investment Decision Tool. Please take a look at the attached file. You should download it first and then open it in your web browser. I have also included a brief user guide for reference.*
>
> *What you now have is a custom, interactive planning tool that translates a portion of real corridor parcels into a live decision-making environment aligned to your four strategic pillars: Housing and Economic Development, Youth and Education, Sustainable Public Realm, and Cultural Arts Ecosystem.*
>
> *Each parcel in the tool reflects the 2026, 2027, and long-term outcomes from your internal strategy framework. Any stakeholder can click a site, assign a pillar, and immediately see the projected impact based on that decision. The intent was to create something that removes the gap between strategy and action and allows both technical and non-technical stakeholders to engage in the same decision-making space.*
>
> *I do want to note that this version requires additional refinement. The spatial positioning of parcels is not yet fully aligned to their exact geographic layout along the corridor. With additional time, this can be calibrated to reflect a more precise spatial relationship. That said, the core system is fully functional. The impact calculator is tied to each parcel's actual square footage, and each site is connected to Google Maps to provide real-world context and visibility.*
>
> *From a broader perspective, this is not simply a visualization. It is a decision-making system designed to reduce the time required to evaluate sites, improve alignment among stakeholders, and create a shared understanding of how individual parcel decisions contribute to corridor and city-level outcomes.*
>
> *To provide context, if this scope were delivered through a traditional planning or design firm, it would typically involve multiple phases, including data analysis, mapping, scenario modeling, and stakeholder facilitation. Engagements of that nature often range from $15,000 to $50,000 or more depending on scope, timeline, and level of customization. This tool was developed through several iterations to align directly with your strategy and to create a usable, decision ready interface within days.*
>
> *Given that this version represents an initial sample and proof of concept, I am valuing this phase at $1,500. This reflects the intent to demonstrate the model, test usability, and establish a foundation for potential expansion, while still recognizing the underlying value of the system being built.*
>
> *If there is interest in continuing this work, I see a strong opportunity to build on this foundation. Future phases could include refining spatial accuracy, expanding across the full corridor, integrating GIS data, and incorporating additional interactive components to support real-time planning, stakeholder engagement, and investment alignment. I would welcome the opportunity to support that next phase as a continued partner.*
>
> *For a foundation making multi-parcel investment decisions, even one misaligned site can result in significant unrealized impact. This tool is intended to reduce that risk while accelerating decision-making and communication.*
>
> *It has been a genuine privilege to work on this. The McNichols corridor represents the kind of thoughtful, place-based investment that I value deeply. I look forward to your feedback and to discussing how this can continue to evolve to support your work.*
>
> *Please let me know if you would like to schedule time to walk through the tool together.*
>
> *Warm regards,*

Message 2 — acceptance, same day

- **From:** Tosha M. Tabron TMTabron@kresge.org
- **To:** christa stoneham
- **Date:** Mon, May 4 [2026], 9:48 PM

> *You are so talented! Thank you. I am showing this to my team tomorrow and will absolutely send you $2,500 for this first phase.*

What this email establishes (facts, for counsel's legal analysis)

1. **Delivery of the working artifact on 2026-05-04** — the complete tool (a single HTML file that runs in a browser) plus a user guide were transmitted to a person outside the project, with instructions to open and use it. No confidentiality language, NDA reference, or restriction on further sharing appears anywhere in the thread.
2. **A commercial transaction on the same day.** The delivery email states a price ("I am valuing this phase at $1,500") for the delivered tool; the recipient accepted the same evening and committed to pay $2,500 "for this first phase." **Counsel should evaluate this as a potential on-sale bar event (35 U.S.C. § 102(a)(1)) in addition to a public-disclosure event** — the sale analysis matters even if the disclosure were somehow deemed non-public. UNVERIFIED: whether/when the $2,500 was actually paid, and under what payment instrument (preserve the record).
3. **Immediate onward dissemination.** The recipient stated she would show the tool to her team the next day (2026-05-05), inside The Kresge Foundation.
4. **The feature scope disclosed on 2026-05-04** (from the email's own description, pending recovery of the actual attachment): - Real corridor parcels (a subset) in an interactive decision environment - Four named investment pillars (wording then: "Housing and Economic Development" / "Youth and Education" / "Sustainable Public Realm" / "Cultural Arts Ecosystem" — vs. today's "Housing & Commercial Development" and "Cultural & Arts Ecosystem" in `McNichols_Kresge_DEMOv2.html:254-273`) - Click-a-site pillar assignment with immediate projected impact - Impact calculator tied to each parcel's actual square footage (the coefficient engine of TECHNICAL-DISCLOSURE.md §1.1-1.2) - Per-parcel "2026, 2027, and long-term outcomes" (matching the y1/y2/lt narrative structure in the current code) - Google Maps link per site - EXPLICITLY NOT YET PRESENT: accurate spatial/geographic positioning ("not yet fully aligned to their exact geographic layout"), GIS data integration (named as a future phase), scenario modeling at professional grade (named as what traditional firms would do). The email does not mention zoning, assessed values, AMI, scenario save/compare, dashboards, or map views — whether any of these were in the May 4 attachment is UNVERIFIED until the attachment is recovered.
5. **Later-added features have later dates.** Everything in this repo's commit history (2026-06-29 onward) — editable pillars, cost overrides, value/tax projection, scenarios and share files, real parcel geometry, building footprints, offline build — post-dates this delivery unless present in the May 4 attachment.

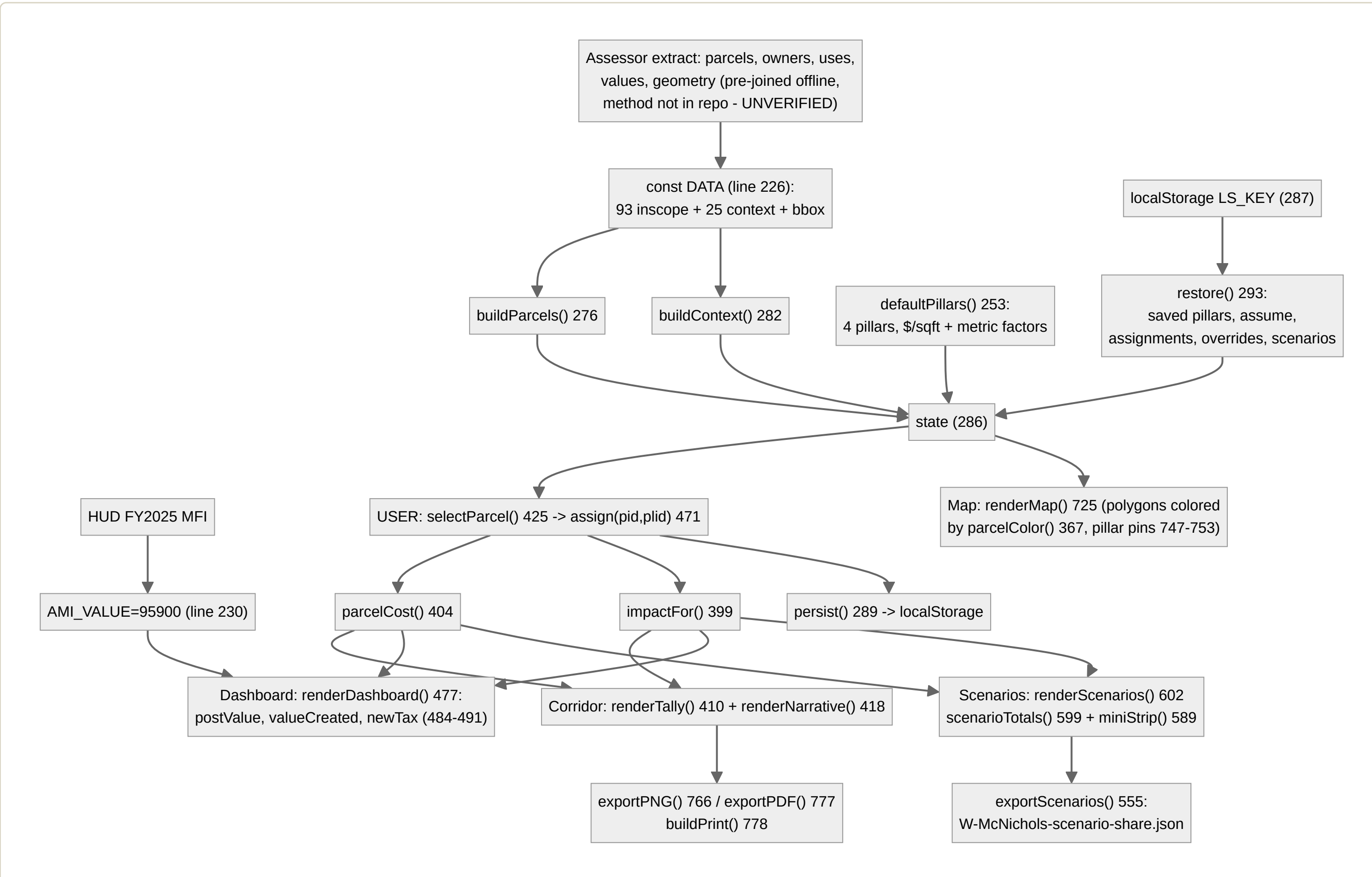
Preservation checklist for the inventor

- [ ] Export the original email (with headers) from Gmail — File > Download message (.eml) — from BOTH accounts (gmail and the "other info email")
- [ ] Preserve the two original attachments exactly as sent (the tool HTML file and the user guide)
- [ ] Preserve the payment record for the $2,500 (date, method)
- [ ] Note any related messages in the same thread (this was "5 of 9" in the Gmail thread view — the other messages in the thread should be reviewed and preserved too)
- [ ] Preserve the ai-planner repository (the codebase the May 4 build came from)

3. Methodology Flowcharts

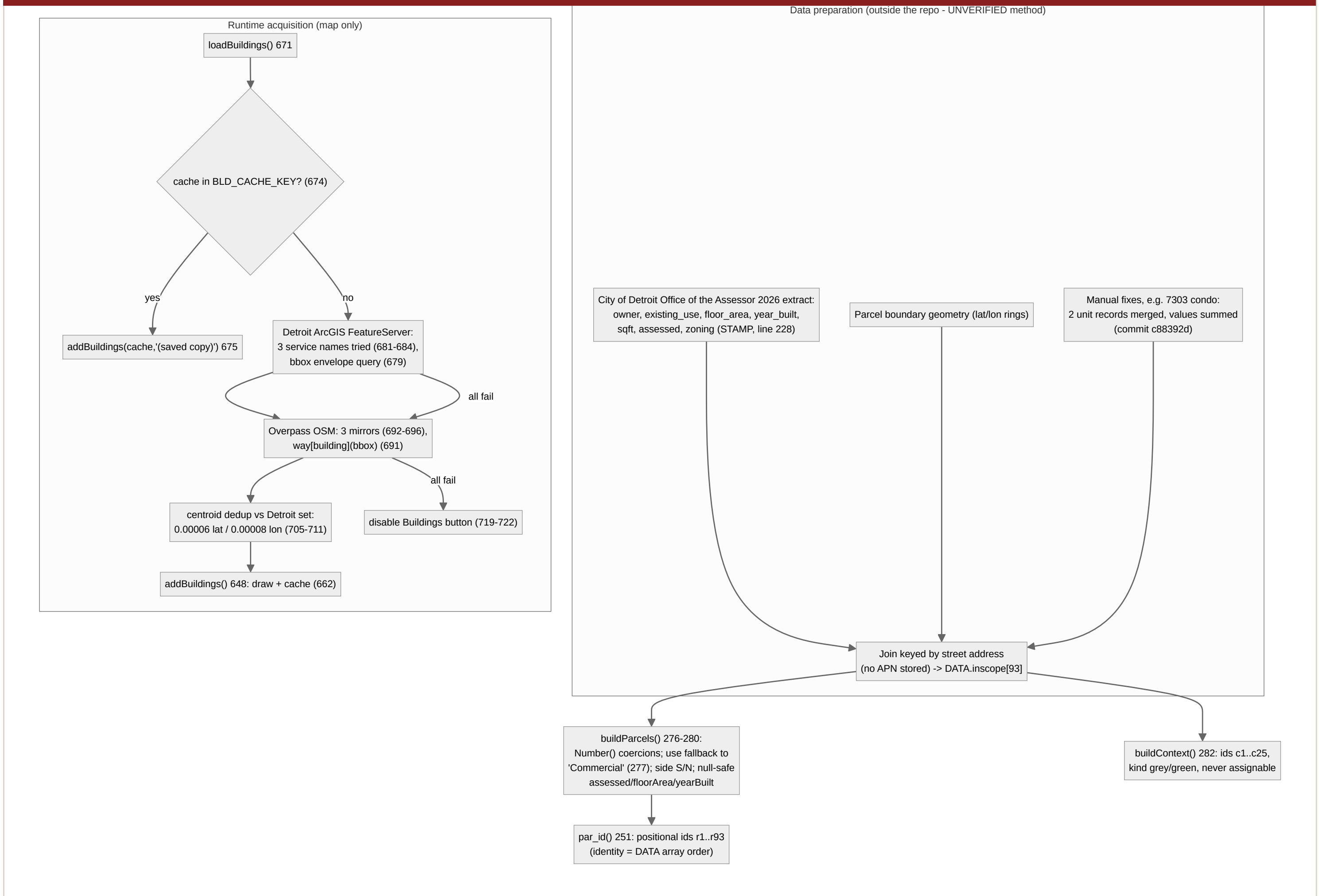
Prepared 2026-08-03. Attorney work product. Every node names the actual function, constant, or file location in `McNichols_Kresge_DEMOv2.html` (line numbers in labels). Companion to TECHNICAL-DISCLOSURE.md.

1. Master flow — raw inputs to the decision output

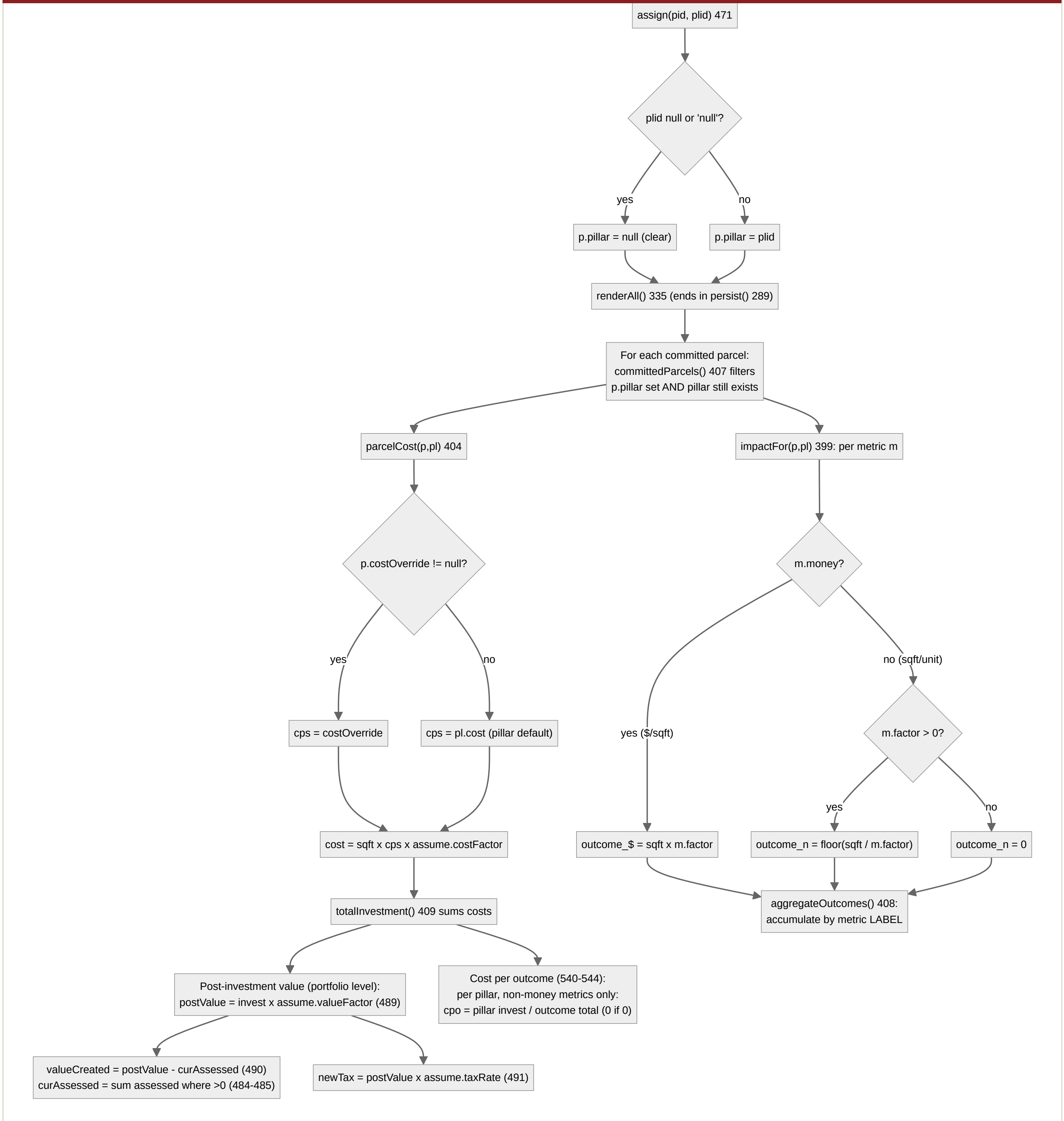


2. Data acquisition and normalization

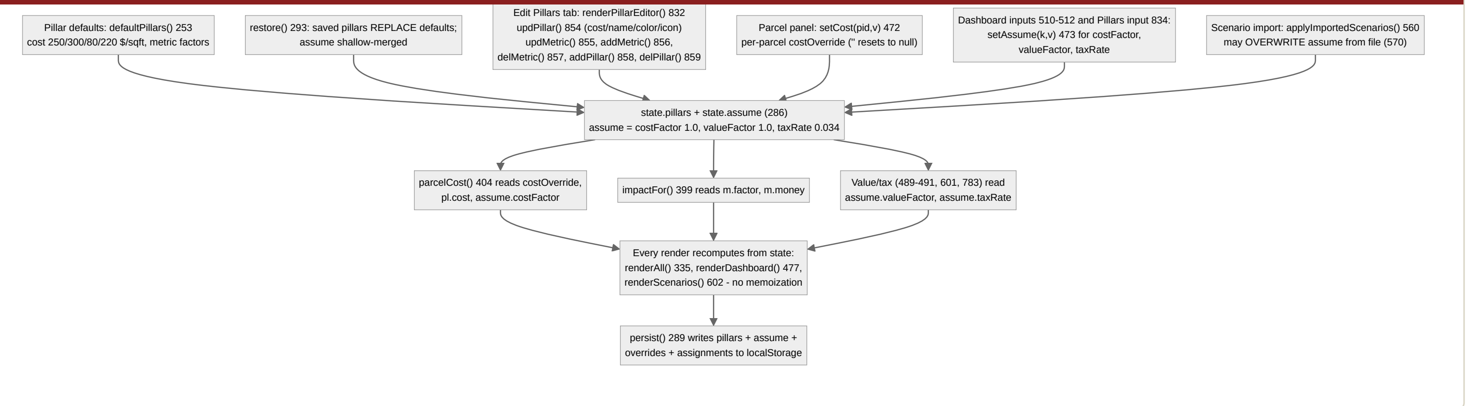
Parcels, zoning, land use, ownership, and valuation are pulled and joined BEFORE the code runs; the file ships with the joined result. AMI is a single embedded constant. Only building footprints are acquired live.



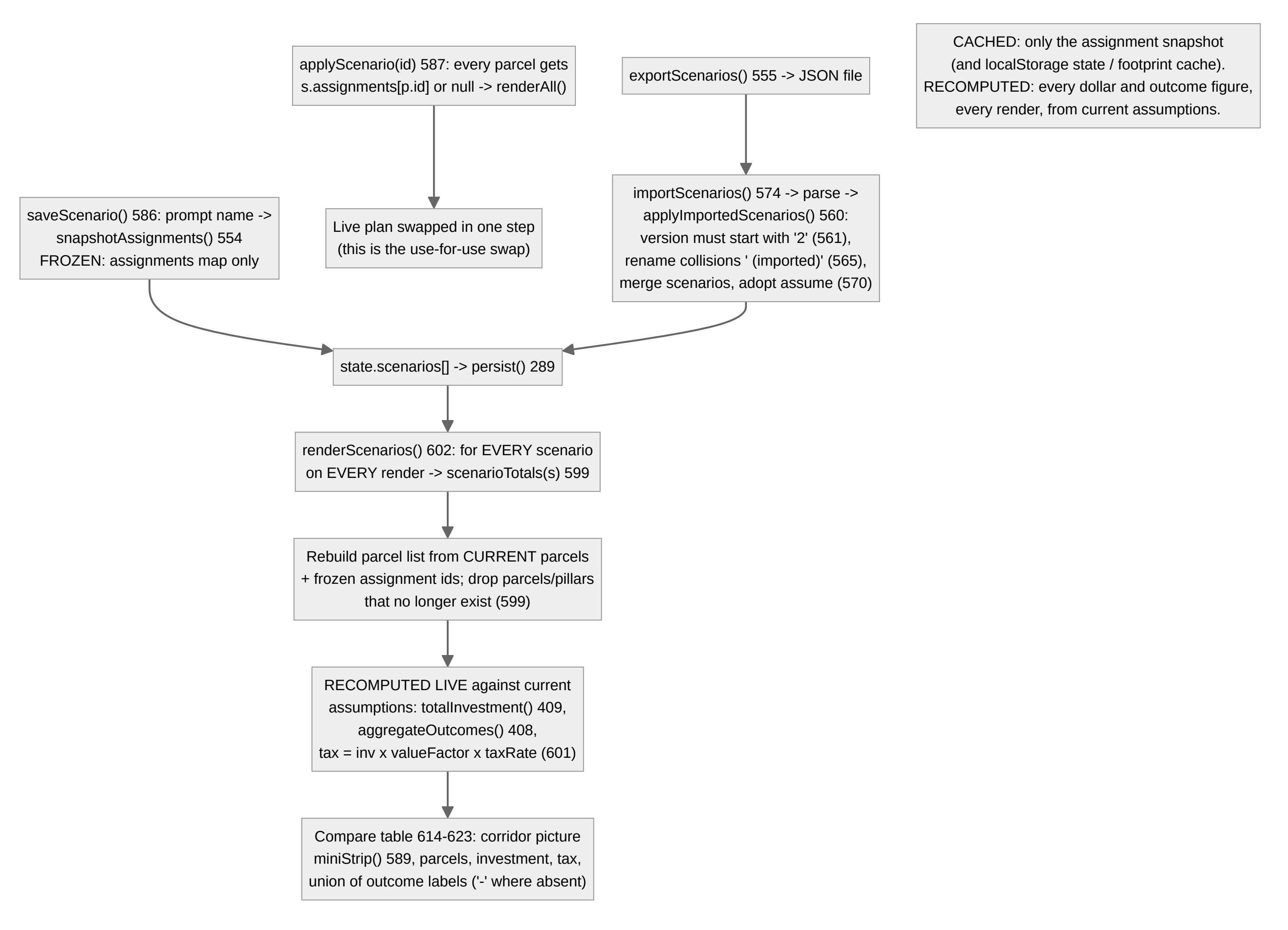
3. Investment pillar logic — every branch



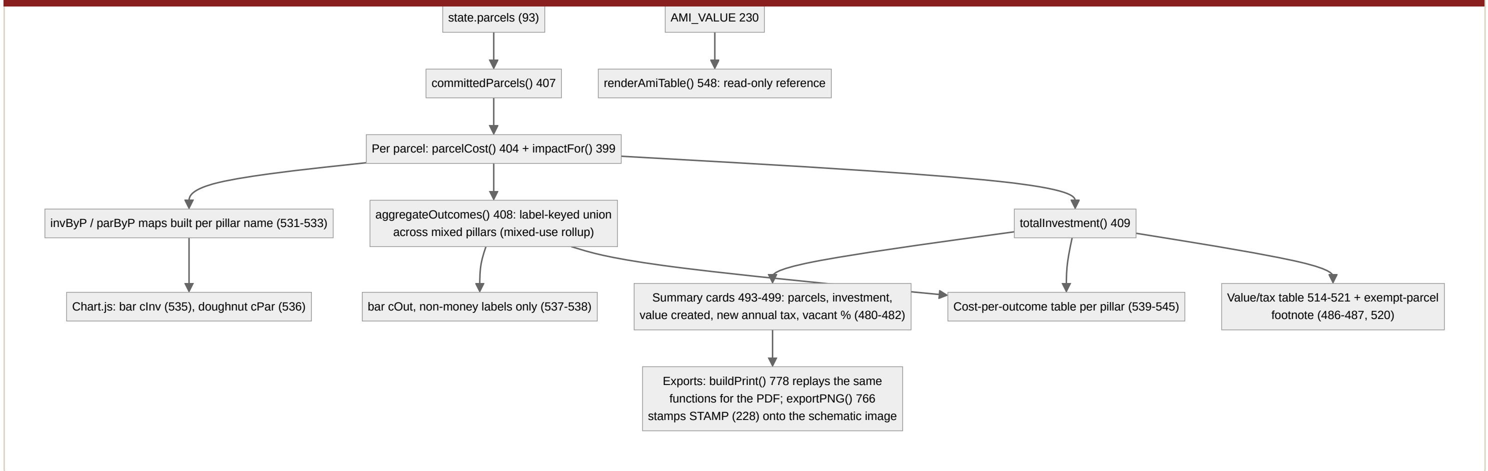
4. Assumption engine — entry, storage, override, propagation



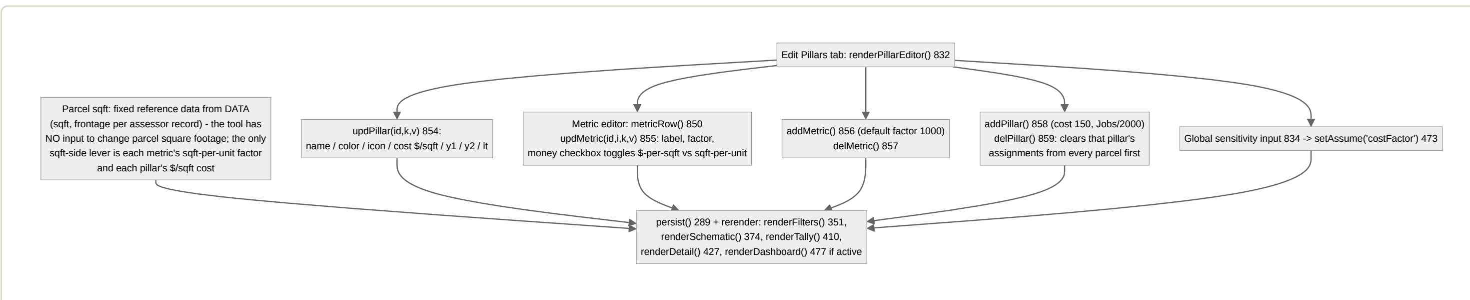
5. Scenario comparison — swap and recompute; cached vs recomputed



6. Portfolio rollup — parcel to dashboard across pillars



7. Configuration flow — pillars and square-footage inputs



Note on diagram 7: the disclosure request asked "how square footage inputs are changed." In this build parcel square footage is immutable assessor reference data (`DATA`, line 226); user-changeable square-footage-related inputs are the metric factors (sqft per unit) and pillar/parcel costs per sqft, as drawn. A proposed-building-sqft input is part of the planned development-modes embodiment (TECHNICAL-DISCLOSURE.md §12.1), not this build.

4. Technical Disclosure

Prepared 2026-08-03. Attorney work product. Written to enable a competent GIS or real estate technologist to rebuild the system from this document alone.

Primary source file: `McNichols_Kresge_DEMOv2.html` (864 lines; "v2", tool version "2.1"). Secondary: `McNichols_Kresge_DEMO.html` ("v1"), `McNichols_Kresge_DEMO_OFFLINE.html` (v2 with libraries inlined). All line numbers below refer to v2 unless stated. All code quotes are verbatim.

0. System overview

A single self-contained HTML file implementing a corridor investment decision tool. It embeds a pre-joined parcel dataset for 93 in-scope parcels plus 25 context parcels on the W. McNichols corridor, Detroit MI (line 226, constant `DATA`, ~45 KB JSON). All computation is client-side JavaScript; there is no server, no build system, and no database. State persists in browser localStorage. Five views (tabs at lines 172-176): Corridor (diagrammatic schematic), Map View (Leaflet GIS), Dashboard (charts + value/tax + AMI), Scenarios (save/compare/share), Edit Pillars (assumption editor).

The core mechanism: a user assigns any parcel to one of N configurable "investment pillars" (use types). Each pillar carries a development cost per square foot and a list of outcome metrics expressed as square-feet-per-unit or dollars-per-square-foot factors. Assignment immediately recomputes per-parcel cost and outcomes, corridor-wide tallies, dashboard aggregates, and a projected post-investment property value and new annual property tax for the assigned set.

1. Formulas

1.1 Per-parcel outcome projection — `impactFor(p, pl)` (lines 399-403)

```
function impactFor(p,pl){const sqft=Number(p.sqft)||0;return pl.metrics.map(m=>{
  let f=Number(m.factor)||0;
  const value=m.money?sqft*f:(f>0?Math.floor(sqft/f):0);
  return {label:m.label,money:m.money,value};
});}
```

For each metric `m` of the assigned pillar `pl`: - If `m.money` is true: **outcome = sqft × factor** — units: dollars (factor is $ per sqft of land per year for revenue metrics; sqft is parcel LAND area). - Else: **outcome = floor(sqft ÷ factor)** — units: whole counts (units, jobs, children, households); factor is sqft of land per one unit of outcome. - A non-money factor of 0 yields 0 (division guarded by `f>0`).

Variables: `p.sqft` = parcel land area in square feet (assessor record, line 278); `m.factor` = pillar metric coefficient (user-editable, see §2.2); `m.money` = boolean flag set per metric (line 853).

Implied assumption: outcomes scale linearly with parcel LAND area, not building area, and are floor-rounded to whole units per parcel (so a parcel smaller than one `factor` produces zero of that outcome).

1.2 Per-parcel cost projection — `parcelCost(p, pl)` (line 404)

```
function parcelCost(p,pl){const cps=(p&&p.costOverride!=null)?Number(p.costOverride):(Number(pl.cost)||0);return (Number(p.sqft)||0)*cps*(state
```

cost = land sqft × cost-per-sqft × global cost factor, in dollars, where cost-per-sqft ($/sqft) resolves in priority order: (1) per-parcel override `p.costOverride` if set (any number, including 0), else (2) the pillar default `pl.cost`. `state.assume.costFactor` is the global sensitivity multiplier (default 1.0).

Implied assumptions: cost scales with LAND area even where a building exists (the UI note at line 833 says "building square footage × cost per square foot" and line 467 says "Estimated cost = {sqft} sqft × cost per sqft" — the words say building sqft in the pillar editor but the CODE uses `p.sqft`, the land area; the parcel-panel text correctly prints the land figure. This wording/code divergence is itself worth noting to counsel). Acquisition cost is not separated from construction cost; the single $/sqft is described in-app as "a planning cost per square foot" (line 240).

1.3 Portfolio investment — `totalInvestment(list)` (line 409)

Sum of `parcelCost` over a parcel list; parcels whose assigned pillar no longer exists contribute 0.

1.4 Portfolio outcome aggregation — `aggregateOutcomes(list)` (line 408)

```
function aggregateOutcomes(list){const t={};list.forEach(p=>{const pl=state.pillars.find(x=>x.id===p.pillar);if(!pl)return;impactFor(p,pl).forE
```

Outcomes accumulate into a map **keyed by metric LABEL string** across all pillars. Two pillars that both define a metric labeled "Jobs" pool into one "Jobs" total even though their factors differ. This label-keyed union is what lets a mixed-use portfolio (housing + arts + parks) roll up into one outcome table without a fixed schema.

1.5 Post-investment property value and tax — `renderDashboard()` (lines 484-491)

```
const valued=list.filter(p=>Number(p.assessed)>0);
const curAssessed=valued.reduce((a,p)=>a+Number(p.assessed),0);
...
const invest=totalInvestment(list);
const postValue=invest*state.assume.valueFactor;
const valueCreated=postValue-curAssessed;
const newTax=postValue*state.assume.taxRate;
```

- **Estimated value after investment = total investment × valueFactor** (dollars). Default valueFactor 1.0 (line 286), i.e. $1 of value per $1 invested, editable (line 511).
- **Estimated value created = postValue − current assessed value of assigned parcels** (only parcels with assessed > 0 count toward `curAssessed`).
- **Estimated new annual property tax = postValue × taxRate** (dollars/year). Default taxRate 0.034 (line 286), labeled "Effective annual tax rate" (line 512).

The on-screen definition (line 508): "Estimated value after investment = total development investment × value factor. Estimated new annual property tax = value after investment × effective tax rate. Planning estimates only, not an appraisal or a tax determination."

Important structural fact: post-investment VALUE is computed at the PORTFOLIO level (from total investment), not per parcel. No per-parcel post-investment value is computed or displayed anywhere in the code. Per-parcel the system computes cost and outcomes only. (See NOVELTY-CANDIDATES.md §1.)

Implied assumptions: value uplift is proportional to spend, independent of use type, location, or existing building; the tax estimate applies the single effective rate to the full post value (not to the increment), and ignores exemption status of the future use; `newTax` is not net of current taxes.

The same formulas recur: - Scenario compare tax (line 601): `tax:inv*state.assume.valueFactor*state.assume.taxRate`. - Print report (lines 780-783): `valueCreated=invest*state.assume.valueFactor-curAssessed`.

1.6 Vacancy KPI (lines 480-482)

```
const vacTotal=state.parcels.filter(p=>p.use==="Vacant").length;
const vacAssigned=state.parcels.filter(p=>p.use==="Vacant"&&p.pillar).length;
const vacPct=vacTotal?Math.round(vacAssigned/vacTotal*100):0;
```

"Vacant land addressed" = assigned vacant parcels ÷ all vacant parcels, %.

1.7 Cost per outcome (lines 540-544)

```
const cost=totalInvestment(pl_list);const agg=aggregateOutcomes(pl_list);
Object.entries(agg).forEach(([l,o])=>{if(o.money)return;const cpo=o.v>0?cost/o.v:0; ... });
```

Per pillar: **cost per outcome = pillar total investment ÷ pillar total of that outcome**, computed only for non-money metrics; 0 when the outcome total is 0. Note the numerator is the pillar's ENTIRE investment, charged in full against EACH outcome type separately (jobs and units each see the whole cost — cost is not allocated across outcome types).

1.8 AMI affordability table — `renderAmiTable()` (lines 548-552)

```
const rows=bands.map(b=>{const inc=ami*b/100;const mo=inc*0.30/12;...});
```

For band b%: **income = AMI_VALUE × b/100**; **affordable monthly housing cost = income × 0.30 ÷ 12**. `AMI_VALUE=95900` dollars, "Detroit-Warren-Livonia, MI (HUD FY2025 Median Family Income)" (line 230). The 30% standard is hard-coded. Read-only reference: AMI does not enter any projection formula in this build.

1.9 Rendering geometry (diagrammatic corridor)

- Parcel cell width px: `widthFor` (372): `Math.max(34, (Number(p.frontage)||40)*1.1)` — 1.1 px per foot of frontage, 34 px floor, 40 ft default frontage.
- Parcel cell height px: `heightFor` (373): `Math.max(46,Math.min(120,s/420))` with `s=sqft||3000` — 1 px per 420 sqft, clamped 46-120.
- Scenario mini-strip cell width px: `miniStrip` (594): `Math.max(2,Math.round((Number(p.frontage)||40)*0.09))` — 0.09 px per foot, 2 px floor.
- Corridor ordering (389): parcels and context sorted by `Number(address)`, split into north/south rows by `side` (391-392).
- Map footprint dedup thresholds (709): centroids within `0.00006`° latitude AND `0.00008`° longitude are duplicates (≈6.7 m × ≈6.6 m at Detroit's latitude).

2. Constants, defaults, coefficients, thresholds

2.1 Global constants

NAME	VALUE	WHERE	MEANING
`TOOL_VERSION`	`"2.1"`	227	version string; gates scenario import (§7)
`STAMP`	"Parcel data: City of Detroit Office of the Assessor, 2026. Boundaries and values are reference data of record. Tool v2.1, Connecting Dots and Strategies."	228	provenance stamp on map, PNG, PDF
`CORRIDOR`, `CITY`	"W. McNichols Corridor", "Detroit, MI"	229	labels
`AMI_VALUE`	95900 ($)	230	HUD FY2025 MFI, Detroit metro
`AMI_METRO`	"Detroit-Warren-Livonia, MI (HUD FY2025 Median Family Income)"	230	label
`LS_KEY`	"cds_mcnichols_demo_v2"	287	localStorage key (v1 uses "cds_mcnichols_demo_v1", v1:281)
`BLD_CACHE_KEY`	"cds_mcnichols_bld_cache_v1"	640	footprint cache key (shared string across v2 and offline builds — cache deliberately interoperates, commit 2a204e3)
tour-done flag	"cds_mcnichols_tour_done"	310, 829	first-run tour gate
`LAND_USES`	9 use→hex color entries	232	category palette
`ZONING_COLORS`	11 district→hex entries; fallback `#7E8AA2`	233-234	zoning palette
expected counts	93 in-scope, 25 context	317-322	console build report sanity checks
AMI bands	[30,50,60,80,100,120] %	549	table rows
affordability share	0.30 of income	550	hard-coded

2.2 Default pillars — `defaultPillars()` (lines 253-274)

PILLAR	COLOR	COST $/SQFT	METRICS (LABEL: FACTOR)
Housing & Commercial Development	#D95448	250	Housing Units: 1200 sqft/unit; Jobs: 2000 sqft/job; Annual Revenue: $4/sqft (money)
Youth & Education	#F1BA4B	300	Children Served: 60 sqft/child; Educator Jobs: 600 sqft/job
Sustainable Public Realm	#47AB4F	80	Park Area (sqft): 1 sqft/sqft; Households Served: 500 sqft/household
Cultural & Arts Ecosystem	#00BCC5	220	Arts Jobs: 800 sqft/job; Studio/Gallery Units: 1000 sqft/unit; Arts Revenue: $5/sqft (money)

Each pillar also carries narrative fields `y1`, `y2`, `lt` (Year 1 / Year 2 outputs, long-term outcomes; lines 256-273) displayed verbatim on assignment.

metric default (`addMetric` , 836): label "New metric", factor 1000.

2.3 Assumption defaults — `state.assume` (line 286)

```
const state={...,assume:{costFactor:1.0,valueFactor:1.0,taxRate:0.034},...};
```

ASSUMPTION	DEFAULT	EDITED AT	PROPAGATION
`costFactor` (global cost sensitivity)	1.0	Dashboard (510) and Edit Pillars (834)	multiplies every `parcelCost` (404)
`valueFactor` (value per $ invested)	1.0	Dashboard (511)	dashboard postValue (489), scenario tax (601), print (783)
`taxRate` (effective annual property tax)	0.034	Dashboard (512)	newTax (491), scenario tax (601)

UNVERIFIED: the empirical basis of every coefficient above (pillar costs, factors, 0.034, 95900). Nothing in the repo documents sources for the pillar coefficients; the UI labels them "planning placeholder" (line 833) and "planning assumption, editable" (line 467). `AMI_VALUE` is attributed to HUD FY2025 but no source file exists in the repo. Calibrated coefficients are an explicitly planned paid deliverable (`McNichols_DEMO_TODO.md:31`).

3. Data model

3.1 Embedded dataset `DATA` (line 226, single JSON literal)

```
DATA = { inscope: ParcelRecord[93], context: ContextRecord[25], bbox: [S,W,N,E] }
bbox = [42.416055,-83.162849,42.417779,-83.139783]
```

`ParcelRecord` fields (observed across all 93 records): `address` (string house number, e.g. "6325"), `owner` (string, uppercase assessor format), `use` (category: Commercial 69, Vacant 20, Governmental 4), `existing_use` (assessor use string of record, e.g. "Store-Retail", "Dry Cleaner", "Gas Station W/Conv Store-Resturant" [sic]; empty for 1+ records), `floor_area` (sqft of building of record, null for 24 records), `year_built` (null for the same 24), `sqft` (land area, range 1,995–48,520), `frontage` (ft), `side` ("N"/"S"), `notes` (string), `zoning` (all 93 are "B2"), `assessed` (dollars; null/0 for 16 records), `rings` (lat/lon polygon array-of-rings; present on all 93).

`ContextRecord` : `address` , `side` , `frontage` , `vacant` (bool), `kind` ("grey"=24, "green"=1 [Marygrove Conservancy]), `name` , `existing_use` , `rings` .

3.2 Runtime entities

- **Parcel** (`buildParcels()` , 276-280): DATA.inscope mapped to `{id, address, owner, use, existingUse, floorArea, yearBuilt, sqft, frontage, side, notes, zoning, assessed, rings, pillar:null, costOverride:null}` . `use` falls back to "Commercial" if not in `LAND_USES` (277).
- **Context** (`buildContext()` , 282-284): ids "c1".."c25"; cannot be assigned; excluded from every tally/export by construction (only `state.parcels` feeds computations).
- **Pillar**: `{id, name, color, icon, cost, metrics[], y1, y2, lt}` .
- **Metric**: `{label, factor, money?}` .
- **Scenario**: `{id, name, date, assignments: {parcelId→pillarId}}` (`saveScenario` , 586; `snapshotAssignments` , 554). Assignments only — cost overrides and pillar definitions are NOT captured in a scenario (noted as a planned change, `McNichols_DEMO_TODO.md:114-115`).
- **state** (286): `{parcels, context, pillars, scenarios, assume, filter, colorMode, selected, view}` .

3.3 Persistence schema — localStorage under `LS_KEY` (persist/restore, 289-300)

```
{a:{parcelId→pillarId}, co:{parcelId→costOverride}, pillars:[...], assume:{...}, scenarios:[...]}
```

Restore merges: saved pillars replace defaults entirely if non-empty (295); assume shallow-merges over defaults (296); assignments/overrides re-attach by parcel id (297); id counters fast-forward past saved ids to avoid collisions (295, 299).

3.4 Scenario share-file schema (`exportScenarios` , 555-559)

```
{version:TOOL_VERSION, scenarios:[...], assume:{...}}   // downloaded as W-McNichols-scenario-share.json
```

- Runtime parcel ids are **sequential** — `par_id()` returns "r1","r2",… (line 251) in the array order of `DATA.inscope` (line 277). Identity is therefore POSITIONAL: it depends on the embedded dataset's ordering staying stable between sessions and between the v2 and offline builds. localStorage assignments (`o.a[p.id]` , 297) and shared scenario files (`s.assignments[pid]` , 587) both key on these positional ids. **Consequence:** reordering, adding, or removing a record in `DATA.inscope` would silently re-map every saved assignment. No assessor parcel number (APN) is stored.
- **Cross-source matching happens at data-preparation time, not runtime.** The assessor attributes (owner, use, value, floor area) and parcel geometry arrive already joined inside `DATA` . The join method is not in the repo — UNVERIFIED; the street address appears to be the working key (records carry only `address` , no APN). Commit c88392d documents a manual identity resolution: a condominium at 7303 had two unit records merged into one parcel with combined geometry and combined assessed value.
- **Building footprints are never joined to parcels.** They are a separate visual overlay layer (non-interactive, `interactive:false` , line 656). Identity between the Detroit and OSM footprint sources is resolved by centroid proximity dedup (§1.9 thresholds, lines 705-711).
- Scenario import collision handling is by scenario NAME: a duplicate name gains the suffix " (imported)" (565).

5. Error, gap, and missing-data handling

GAP	BEHAVIOR	WHERE
No/zero assessed value (16 of 93 parcels)	Detail shows "Exempt / no value of record"; excluded from `curAssessed` sums; counted and disclosed in dashboard footnote ("{exemptAll} of {N} parcels are tax exempt or carry no assessed value…") and print report	442, 484-487, 520, 780-781, 797
No building of record (24 of 93)	Detail shows "No building of record"; floor area plays no computational role in this build	448
No owner	"Not available"	445
No zoning	"Not available" (in data, all 93 have B2)	452
Unknown land-use category	coerced to "Commercial"	277
Missing geometry	build report console warning; parcel simply not drawn on map; still fully functional in schematic/dashboard	320-321, 739, 757
Assigned pillar later deleted	assignment cleared on delete (859); defensive filter drops orphans from tallies (`committedParcels` , 407; `aggregateOutcomes` , 408; `scenarioTotals` , 599)	
Metric factor 0 / negative sqft	outcome 0 (guards at 399-401); sqft coerced `Number(...)||0` throughout	
localStorage unavailable	all persist/restore/cache wrapped in try/catch, silently degrades to in-memory session	289-300, 662, 674-676
Chart.js missing (offline first run)	dashboard renders tables/cards without charts (`typeof Chart==="undefined"` guard)	529
html2canvas missing	PNG export alerts and points to PDF path	766
All footprint sources fail	console warn; Buildings button disabled and relabeled "Buildings unavailable"; note updated	719-722, 645-647
Malformed scenario import	JSON parse alert; version/shape validation returns −1 → "not a scenarios export from this tool (version 2)"	560-561, 578-580

6. Computation order and dependencies

1. `init()` (302): `buildParcels()` → `buildContext()` → `defaultPillars()` → stamp injection → `restore()` (localStorage overlays defaults) → `renderAll()` → `buildReport()` (console diagnostics) → first-run tour.
2. `renderAll()` (335) = renderLegend → renderFilters → renderSchematic → renderTally → renderNarrative → renderDetail → **persist()**. Every mutation funnels through a render call that ends in persist, so the stored state is always current; there is no dirty-tracking.
3. Mutations: `assign` (471), `setCost` (472), `setAssume` (473), pillar edits (854-859) — each calls `renderAll()` (or targeted renders) and, when the affected view is active, `renderMap()` / `renderDashboard()` / `renderPillarEditor()` .
4. **Everything is recomputed from scratch on every render; nothing is cached or memoized.** Scenario compare recomputes `scenarioTotals` per scenario per render (599-601) against CURRENT pillars/assumptions — so a saved scenario's dollars shift if assumptions change after saving (assignments are the only frozen part). The only caches in the system are localStorage state (§3.3) and the footprint cache (§8); Chart.js instances are destroyed and rebuilt each dashboard render (`destroyCharts` , 476, 530).
5. Map lifecycle: `ensureMap()` builds the Leaflet map once per session (629), fits bounds from all polygon points, else `DATA.bbox` (635-636), then `loadBuildings()` once (`bldTried` latch, 672).

- Save (586): name prompt → snapshot of current assignments only.
- Load (587): overwrites every parcel's pillar from the scenario (parcels absent from the scenario get null).
- Compare (602-624): table over ALL saved scenarios — corridor mini-strip picture (`miniStrip`, 589-598), parcel count, investment, est. new annual property tax, then the union of outcome labels across scenarios with "-" for absent metrics (615-622).
- Share (555-585): JSON download / file-picker import; import requires `d.version` to be a string starting with "2" and `d.scenarios` to be an array (561); imports merge (never replace), rename on name collision, and may overwrite `assume` from the file (570).

8. Aerial imagery and building footprint handling

- Base layers (630-632): OSM raster tiles and Esri World_Imagery tiles; satellite on by default (`satOn=true`, 628), toggle swaps layers (724).
- Parcel polygons: in-scope drawn from `rings` at fillOpacity .78 colored by pillar/use/zoning (`parcelColor`, 367-371; 738-746); context in grey/green at .4/.5 (728-734); selected parcel gets dark 3px outline (742). Assigned parcels get a non-interactive emoji `divIcon` marker at the vertex-average centroid (736-737, 747-753).
- Footprints (`loadBuildings`, 671-723), order of precedence: 1. localStorage cache (`BLD_CACHE_KEY`) — used verbatim, labeled "(saved copy)", never refreshed once present (675). 2. City of Detroit ArcGIS FeatureServer — three candidate service names tried in order (`Buildings`, `Building_Footprints`, `building_footprints`, 681-684) with an envelope query built from `DATA.bbox` returning GeoJSON (679); first success wins. 3. OSM Overpass — three mirrors tried in order (693-695) with query `way["building"](bbox); out geom;` (691); results converted to GeoJSON Polygons (701). If Detroit data already loaded, OSM SUPPLEMENTS it: footprints whose centroid falls within the §1.9 thresholds of an existing one are dropped as duplicates, the rest are added (703-713). 4. On any success the merged set is cached with its source label (662); on total failure the layer is disabled with user-visible notice (719-722).
- Rendering (648-658): GeoJSON Polygon/MultiPolygon → Leaflet polygons, ink outline at .75 opacity, .07 fill, non-interactive overlay above parcels.
- The offline build performs identically (same code); its inlined libraries remove the CDN dependency but tiles and first footprint fetch still require internet; the shared `BLD_CACHE_KEY` lets a cache captured in one build serve the other (commit 2a204e3).

9. Zoning and land-use ingestion and interpretation

- Zoning arrives as a per-parcel district string of record (all 93 = "B2"); displayed read-only with "reference" badge (452), colored via `ZONING_COLORS` map with grey fallback (233-234), togglable color mode on schematic and map (359-363), legend states "All in-scope parcels are zoned B2 (local business). Reference data." (344). **Zoning is never interpreted or enforced** — no rule maps zoning to permitted pillars, density, or envelope (a B2 setback/envelope analysis is an explicitly deferred item, `McNichols_DEMO_TODO.md:27-28`).
- Land use is two-level: broad category `use` (drives color, vacancy logic, filters) and assessor `existing_use` string of record (display-only, 447). The category vocabulary and its 9 colors are fixed at line 232.
- The abandoned readiness checklist (see §11) contained the only zoning-aware logic ever present: a human-checked "Policy and zoning ready" flag.

10. AMI data integration and what it drives

`AMI_VALUE=95900` (line 230) drives exactly one artifact: the read-only dashboard reference table (§1.8) of income bands and 30%-of-income monthly housing costs (523-526, 548-552). It does NOT parameterize unit counts, rents, eligibility, or any projection. An "AMI-aware housing pillar" that ties unit counts and rents to AMI bands is a documented planned embodiment (`McNichols_DEMO_TODO.md:25-26`).

11. Alternative approaches visible in the code and history

Preserved for claim breadth; none of this should be treated as deleted subject matter.

1. **Readiness scoring (implemented, then removed 2026-06-29, commit 336bbf3).** The original build (`git show 913a55a:McNichols_Kresge_DEMO.html`) scored each parcel on five boolean risk dimensions — funding / policy+zoning / stakeholder / environmental+title / timeline (`const READINESS`, 913a55a line 218) — with `rdScore(p)` counting checked flags, a percentage badge, and a red/amber/green threshold color: `rdColor(pct){return pct>=70?"#47AB4F":pct>=40?"#F1BA4B":"#D95448";}`. Readiness state persisted per parcel in localStorage. This is a parallel qualitative-risk axis alongside the quantitative projections — an alternative embodiment of decision support that could return in Tier 2/3.

no per-parcel override, no sensitivity factor. The override + global factor chain is a second generation refinement (336bbf3, edeee1a).

3. **Two coexisting builds of the same version** (online v2 vs OFFLINE with inlined libraries) — alternative packaging embodiments; plus the v1 file retained in-repo with a separate localStorage namespace ("…_v1" vs "…_v2"), demonstrating side-by-side version isolation.
4. **Three-candidate ArcGIS service-name fallback and three Overpass mirrors** (681-684, 692-696) — redundant-source acquisition strategy.
5. **Unused/vestigial code:** `findCtx` /context detail path duplicated between corridor and map panels (427, 763); `p.notes` displayed but sparsely populated; `DATA.bbox` fallback for fitBounds only used if no polygon exists (636, never in practice since all 93 have rings). `heightFor` uses a hard default of 3000 sqft when data absent (373, never in practice).
6. **No feature flags and no commented-out logic** exist in v2 (verified by read-through); the alternatives live in git history and in the TODO file rather than dead code.
7. **Wording/code divergence** on cost basis (building vs land sqft), §1.2 — evidence that a building-area cost basis is a contemplated variant; the development-modes plan below makes that explicit.

12. NOT YET BUILT — intended embodiments

Documented in `McNichols_DEMO_TODO.md` (committed 2026-07-03, extended 2026-07-06 and 2026-07-13) and, where noted, in conversation-derived roadmap text inside that file. Each is described against the existing architecture so a provisional can cover it with specificity.

1. **Development modes (rehab / new construction / demolition+new)** (TODO:21-24). A per-parcel `mode` field beside `costOverride`; `parcelCost` switches basis: rehab = existing `floorArea` × rehab $/sqft; new = proposed building sqft × new-construction $/sqft; demo+new adds a demolition line. The data foundation (floor_area, year_built per parcel) already ships in `DATA` (§3.1).
2. **AMI-aware housing pillar** (TODO:25-26). Housing metrics parameterized by AMI band: unit counts split across bands (30–120% of `AMI_VALUE`), band rents derived from the §1.8 affordability formula, feeding revenue and subsidy-gap lines in the pro forma below.
3. **Pro forma gap analysis** (TODO:32, "Preliminary pro formas; capital stack"). Per scenario: development cost (existing `totalInvestment`), stabilized income (money-type metrics per §1.1, e.g. Annual Revenue), supportable debt from income, equity, and the **gap = cost − (debt + equity)**; rendered as a dashboard card and per-scenario compare row using the same recompute-on-render pattern (§6.4). Assumption entry follows the existing `state.assume` + `setAssume` propagation path (§2.3).
4. **Funding fill-in (pipeline) tracking** (TODO:32, "funding pipeline status"). A per-scenario `sources[]` list ({name, program, amount, status: identified/applied/committed}) whose committed sum draws down the pro-forma gap; status rolls up to the compare table so funders see which plan is closest to fully funded. Persisted in the same localStorage/share-file schema (§3.3-3.4, with the share format's version gate bumped).
5. **Environmental screening, three rings** (TODO:39-70, design approved 2026-07-06). Ring 0: `ENV_USE_FLAGS` mapping assessor use codes already in `existing_use` (dry cleaner, gas station, service/repair, car wash) to plain-language Phase-I-ESA screening flags in the parcel panel. Ring 1: a live EGLE Part 201/Part 213 point layer cloning the `loadBuildings()` pattern — bbox query, warning-style markers, localStorage cache with "(saved copy)" fallback, ~500 ft proximity lines in the parcel panel. Ring 2 (service, not software): records pulls, Phase I coordination, brownfield funding pathway.
6. **Architectural visualization, three altitudes** (TODO:72-107, design 2026-07-13). (a) Map "proposed" mode: setback-respecting concept-massing rectangles tinted by pillar over existing footprints, before/after toggle; (b) street elevation tab: diagrammatic corridor wall from frontage + floor-area of record vs proposed massing in pillar colors; (c) per-site renderings produced outside the tool and attached to parcel detail / funder packet. All three read the same pillar assignments (single source of truth).
7. **Remaining TODO items:** clear-all-assignments control (TODO:5-8); tour updates (9-14); assessor vintage confirmation (15-16); calibrated projection coefficients (31); decision log + weighted criteria (33); phasing view of Year 1/2/3 capital needs and sensitivity stress tests (34); multi-page funder packet PDF export (35); study-grade millage-based tax modeling replacing the single effective rate (36); additional corridors (37); hosted deployment (110-111); multi-user shared workspace with a shared scenario database (112-113); embedding pillar definitions in the scenario share file so edited assumptions travel with scenarios (114-115).
8. **B2 buildable-envelope analysis** (TODO:27-28): zoning-interpretation layer computing setbacks/envelope per parcel — the first rule-based use of the zoning field (§9).

UNVERIFIED items in this section: none are implemented; all descriptions of HOW they would work are architecture-consistent designs recorded in the repo's TODO file (quoted locations given), except the pro forma debt-sizing detail in item 3, which is the standard mechanism implied by "pro formas; capital stack" but not spelled out in the repo.

5. Data Inventory

Prepared 2026-08-03. Attorney work product. Sources are listed as evidenced in the repository only; no outside research was performed (per instructions). File references are to `McNichols_Kresge_DEMOv2.html` (v2) unless noted; the OFFLINE build contains the same data-source code with libraries inlined.

1. External data source table

			METHOD	TOUCHES IT		METHOD	IN REPO?
1	City of Detroit Office of the Assessor (2026 extract)	Parcel ownership, existing use of record, land sqft, frontage, building floor area, year built, assessed value, zoning district, parcel boundary rings for 93 in-scope + 25 context parcels	Manual/offline extraction and join, method not in repo (UNVERIFIED)	Embedded as `const DATA`, v2 line 226; consumed by `buildParcels()` 276, `buildContext()` 282; attributed by `STAMP` line 228	JSON embedded in HTML	None — static snapshot baked into the file; TODO:15-16 flags confirming the vintage year	No. RESEARCH REQUIRED
2	City of Detroit open data, ArcGIS FeatureServer (`services2.arcgis.com/qvkbeam7Wirps6zC`)	Building footprint polygons within corridor bbox	Live REST API query at runtime (`fetch`, GeoJSON out), 3 candidate service names	`loadBuildings()` 671-690 (`arcgisCandidates` 680-684, envelope query 679)	GeoJSON	Fetched once per browser, then localStorage-cached forever under `BLD_CACHE_KEY` (640, 662, 674-675)	No. RESEARCH REQUIRED
3	OpenStreetMap via Overpass API (3 mirrors: overpass-api.de, overpass.kumi.systems, maps.mail.ru)	Supplemental building footprints (`way["building"]`)	Live HTTP query at runtime	`loadBuildings()` 691-717 (`overpassMirrors` 692-696, dedup 705-711)	Overpass JSON → converted to GeoJSON (701)	Same cache as #2	No, beyond the © OpenStreetMap attribution string on the tile layer (631). RESEARCH REQUIRED (ODbL)
4	OpenStreetMap raster tile server (`tile.openstreetmap.org`)	Street basemap tiles	Live tile requests via Leaflet	`ensureMap()` 631	PNG tiles	Live per pan/zoom; not cached by the app	Attribution "© OpenStreetMap" set at 631; usage policy not in repo. RESEARCH REQUIRED
5	Esri ArcGIS Online World_Imagery (`server.arcgisonline.com`)	Satellite/aerial basemap tiles (default base layer)	Live tile requests via Leaflet	`ensureMap()` 632, `toggleSat()` 724	JPEG/PNG tiles	Live per pan/zoom	Attribution "Esri" set at 632; license terms not in repo. RESEARCH REQUIRED
6	HUD FY2025 Median Family Income, Detroit-Warren-Livonia MI	Single AMI dollar figure used for the affordability reference table	Manually transcribed constant	`AMI_VALUE=95900`, line 230; `renderAmiTable()` 548	Hard-coded number	Manual re-entry ("published yearly by HUD", TIPS.ami line 243)	No. RESEARCH REQUIRED (US federal data, expected public domain — verify)
7	Google Fonts (fonts.googleapis.com / fonts.gstatic.com)	Fraunces and Inter typefaces	Live CSS/font fetch	v2 lines 8-10	WOFF2	Live each load (v2); OFFLINE build note: font links remain remote in the offline file	No. RESEARCH REQUIRED (fonts themselves are OFL — verify)
8	unpkg.com CDN	Leaflet 1.9.4 JS + CSS	Live script/style fetch (v2 only; inlined in OFFLINE)	v2 lines 7, 222	JS/CSS	Pinned version	License header embedded in OFFLINE build (BSD-2-Clause)
9	cdnjs.cloudflare.com CDN	Chart.js 4.4.1, html2canvas 1.4.1	Live script fetch (v2 only; inlined in OFFLINE)	v2 lines 223-224	JS	Pinned versions	License headers embedded in OFFLINE build (MIT)
10	Google Maps (www.google.com/maps/search)	Outbound "Open in Google Maps" links per parcel (no data ingested)	User-clicked hyperlink only	`detailHTML()` 455, `ctxDetailHTML()` 436	n/a	n/a	n/a (outbound link; no API use)

exists; all live access is public HTTPS and the servers.

2. Redistribution and resale analysis (repo evidence only)

For each source: does the REPO contain license terms, attribution requirements, or commercial-resale restrictions?

1. **Detroit assessor extract (embedded `DATA`).** The repo contains no license or terms. It does contain self-imposed attribution/provenance: the `STAMP` string (line 228) credits "City of Detroit Office of the Assessor, 2026" and is rendered on the map, PNG export, and PDF export. The embedded dataset includes personal names of parcel owners — note for counsel: these are public records, but they ARE redistributed inside the HTML file itself (anyone given the file receives all 93 owner names). **RESEARCH REQUIRED** (City of Detroit open data license, assessor data terms).
2. **Detroit ArcGIS building footprints.** No terms in repo. Footprint geometries are cached into the user's browser and, per commit 2a204e3, persist across builds; they are not committed to the repo itself. **RESEARCH REQUIRED.**
3. **OpenStreetMap data (Overpass footprints).** No terms in repo beyond the "© OpenStreetMap" attribution on the tile layer. OSM data is ODbL; ODbL is a share-alike DATA license — whether caching/merging OSM footprints with city data creates a "derivative database" with share-alike obligations is a real question for counsel. **RESEARCH REQUIRED (ODbL §4.4 derivative database analysis).**
4. **OSM tile server.** Attribution present (631). The public OSM tile server has a usage policy (heavy/commercial use discouraged) not present in the repo. Relevant if the tool is hosted commercially at scale. **RESEARCH REQUIRED.**
5. **Esri World_Imagery tiles.** Attribution string "Esri" present (632) but Esri's terms for World_Imagery generally require more than a bare "Esri" credit and may require an ArcGIS subscription for commercial apps. Nothing in repo. **RESEARCH REQUIRED — this is the most likely source to carry a commercial-use restriction.**
6. **HUD AMI figure.** Single transcribed number with in-app attribution (lines 230, 243, 524). **RESEARCH REQUIRED** (expected unrestricted as US government work — verify).
7. **Google Fonts.** No terms in repo. **RESEARCH REQUIRED** (Fraunces and Inter are expected SIL OFL 1.1 — verify; OFL permits commercial use).
8. **Leaflet / Chart.js / html2canvas.** The OFFLINE build embeds the libraries WITH their license headers — this is in-repo evidence: Leaflet BSD-2-Clause ("© 2010-2023 Volodymyr Agafonkin, © 2010-2011 CloudMade", header in `McNichols_Kresge_DEMO_OFFLINE.html`), Chart.js MIT (chartjs.org header), html2canvas MIT (hertzen.com header). All three permit commercial redistribution with notice preservation. See LICENSE-RISK.md.

RESEARCH REQUIRED list (terms not discoverable from the repo)

1. City of Detroit assessor data / open data portal license (sources 1, 2)
2. OpenStreetMap ODbL obligations for merged + cached footprints (source 3)
3. OSM tile server usage policy for hosted commercial use (source 4)
4. Esri World_Imagery commercial licensing (source 5)
5. HUD income-limit data terms (source 6)
6. Google Fonts / Fraunces / Inter font licenses (source 7)
7. unpkg and cdnjs CDN terms of service for production commercial apps (sources 8, 9)

6. Novelty Candidates

Prepared 2026-08-03. Attorney work product. Written for a professional patent searcher who has never seen this product. All code references are to `McNichols_Kresge_DEMOv2.html` (v2). "Core" = load-bearing to the product's value proposition; "peripheral" = supporting.

The system in one sentence: a self-contained, client-side corridor investment decision instrument that binds a pre-joined parcel-of-record dataset to a set of user-configurable "investment pillars" (use types carrying cost-per-sqft and outcome-per-sqft coefficients), recomputing per-parcel costs and outcomes, portfolio rollups, and a projected post-investment property value and tax yield on every assignment, with savable, shareable, visually-compared scenarios.

The inventor's four believed distinguishing elements — verdicts first

A. "Projected post-investment property value from a selected use type at the individual parcel level"

The code does NOT support this belief as stated. Post-investment value is computed at the PORTFOLIO level only: `postValue = totalInvestment(list) × assume.valueFactor` (lines 488-489), where `list` is all assigned parcels together. No per-parcel post-investment value is computed, stored, or displayed anywhere in the build. What IS per-parcel: cost (`parcelCost`, 404) and outcomes (`impactFor`, 399), both driven by the selected use type; and the per-parcel current assessed value of record is carried (line 279) and subtracted in aggregate (490). Because value is linear in investment (valueFactor × cost) and cost is per-parcel, a per-parcel value figure is arithmetically implied by the existing formulas — but it is an intended embodiment, not an implemented one. **Recommendation: claim the per-parcel value projection as a described embodiment in the provisional (it follows directly from parcelCost × valueFactor against the parcel's assessed baseline) and claim the implemented portfolio-level value/tax projection as the built embodiment.** Do not represent per-parcel value output as current functionality.

B. "Holding multiple investment pillars against the same parcel set and switching between them live"

Supported. Two mechanisms implement this: (1) per-parcel live reassignment among N pillars with immediate full recompute — `assign()` (471) → `renderAll()` (335), with the pillar set itself user-definable at runtime (`addPillar` / `updPillar` / `delPillar`, 854-859); (2) whole-plan swapping — `applyScenario()` (587) replaces the entire assignment map in one action, and `scenarioTotals()` (599) re-prices every saved plan against CURRENT assumptions on every render, so alternatives stay comparable as assumptions move. Core.

C. "Aggregating parcel-level projections into a portfolio view across mixed use types"

Supported. The label-keyed outcome union (`aggregateOutcomes`, 408) is the specific mechanism: each pillar carries its own arbitrary metric schema (user-editable labels/factors), and portfolio rollup accumulates by metric label across heterogeneous pillars without a fixed outcome taxonomy — housing units, jobs, park sqft, and arts revenue coexist in one tally, dashboard, and scenario-compare table (410-417, 477-545, 614-623). Core.

D. "Consolidating GIS, zoning, land use, ownership, valuation, and AMI into one decision surface rather than separate reports"

Supported, with one honest qualifier. All six data families are present on one surface: parcel polygons on satellite + live building footprints (GIS, 629-758), zoning of record with color mode (233, 359-363, 452), two-level land use (232, 277, 446-447), ownership (445), assessed valuation feeding the value/tax card (453, 484-491), and AMI (230, 548-552). The qualifier: **AMI is a read-only reference table; it does not parameterize any computation in this build** (the AMI-aware housing pillar is a documented planned embodiment, `McNichols_DEMO_TODO.md:25-26`). Also note zoning is displayed, never interpreted (no rule engine). The consolidation claim is honest as "one decision surface"; it would overreach as "AMI-driven underwriting." Core.

Full candidate list

1. Coefficient-driven use-type assignment engine (built)

- **Claim-like statement:** A system that associates each of a plurality of user-configurable use types with a cost-per-unit-area coefficient and a set of outcome coefficients, and, upon assignment of a land parcel of record to a use type, computes a development cost and a plurality of outcome corridor, and portfolio displays.

- **Technical steps:** parcel record with `sqft` of record (226, 278) → assignment (471) → `parcelCost` = sqft × ($/sqft override ?? pillar default) × global sensitivity (404) → `impactFor` = sqft×factor ($) or floor(sqft/factor) (counts) (399-403) → synchronous re-render of every view
- persistence (335, 289).
- **Files:** `McNichols_Kresge_DEMOv2.html` (all); same engine in v1 and OFFLINE.
- **Conventional alternative:** a spreadsheet pro forma per site, or GIS suitability scoring that ranks parcels but does not price a chosen use; parcel viewers (Regrid, Landgrid, county GIS) display records without projection.
- **Core.**

2. Live multi-pillar scenario switching re-priced against current assumptions (built)

- **Claim-like statement:** A method of holding a plurality of saved assignment maps over a common parcel set and, on each display cycle, re-computing every saved map's cost, outcome, and tax projections against the currently prevailing assumption set, such that saved alternatives remain commensurable after any assumption change.
- **Technical steps:** frozen assignment snapshot (554, 586) → per-render rebuild of a virtual parcel list per scenario (599) → same cost/outcome/tax functions as the live plan (408-409, 601) → side-by-side compare table with outcome-label union (614-623) → one-step whole-plan swap (587).
- **Files:** v2 lines 554-626.
- **Conventional alternative:** static scenario reports frozen at authoring time; BI dashboards that snapshot numbers, not assignments.
- **Core.**

3. Label-keyed heterogeneous outcome rollup (built)

- **Claim-like statement:** Aggregating outcome projections from parcels assigned to differently-schema'd use types by accumulating values under their textual metric labels, whereby use types with disjoint or overlapping outcome vocabularies roll up into a single portfolio table without a predefined outcome taxonomy.
- **Technical steps:** per-pillar arbitrary metric arrays (255-273, editable 850-857) → `aggregateOutcomes` label-map accumulation (408) → dashboard charts/tables filter by money flag (537-545) → scenario compare unions labels across scenarios with "-" fill (615-622).
- **Files:** v2 lines 408, 537-545, 615-622.
- **Conventional alternative:** fixed KPI schemas (jobs/units columns defined in advance) or manual consolidation across separate models.
- **Core** (it is what makes mixed-use portfolios roll up).

4. Portfolio value-creation and tax-yield projection from assignment state (built)

- **Claim-like statement:** Computing, from an assignment state over parcels of record, an estimated post-investment property value as a scalar multiple of aggregate projected development cost, an estimated value created as that figure net of the parcels' assessed values of record (excluding exempt parcels with disclosure of the exclusion count), and an estimated new annual property tax as the post-investment value times an effective rate, each factor being user-adjustable with immediate recomputation.
- **Technical steps:** lines 484-491 (dashboard), 601 (per-scenario), 780-783 (print); exempt handling 442, 486-487, 520; assumption inputs 510-512 via `setAssume` (473).
- **Files:** v2 lines 442, 473, 484-491, 508-521, 601, 780-783.
- **Conventional alternative:** fiscal-impact studies produced offline by consultants; TIF projection spreadsheets.
- **Core.**

5. Frontage-proportional diagrammatic corridor with scenario "corridor picture" (built)

- **Claim-like statement:** Rendering a street corridor as two facing rows of parcel cells ordered by address and dimensioned from each parcel's recorded frontage and land area, colored by assignment state, and rendering a miniaturized strip of the same encoding as a per-scenario fingerprint in a comparison table.
- **Technical steps:** `widthFor` 1.1 px/ft (372), `heightFor` sqft/420 clamped (373), address-sort + side split (389-392), road band (393-396); miniature: 0.09 px/ft, 2 px floor (589-598) reused in scenario list and compare row (611, 618).
- **Files:** v2 lines 367-397, 589-598.
- **Conventional alternative:** map choropleths or tabular scenario diffs; no schematic elevation-style strip.
- **Core-adjacent** (signature visualization; also the seed of the planned street-elevation embodiment).

6. Multi-source building-footprint acquisition with proximity dedup and offline-persistent cache (built)

- **Claim-like statement:** Acquiring building footprints for a bounding box by querying a municipal feature service under multiple candidate service names, supplementing with crowdsharing the storage key, render footprints without network access.

- **Technical steps:** lines 671-723 (fallback chain), 705-711 (0.00006°/ 0.00008° dedup), 640+662+674-675 (cache), commit 2a204e3 (cross-build persistence).
- **Files:** v2 lines 640-723; `McNichols_Kresge_DEMO_OFFLINE.html` (same code).
- **Conventional alternative:** single-source GIS layer requiring live server.
- **Peripheral** (robustness engineering; possibly claimable as a dependent limitation, not standalone).

7. Versioned plain-file scenario interchange (built)

- **Claim-like statement:** Sharing decision scenarios between users of a server-less tool via a downloaded JSON artifact carrying a tool-version gate, scenario assignment maps, and the author's assumption set, imported by merge with collision renaming.
- **Technical steps:** lines 555-585; version prefix check (561), rename " (imported)" (565), assumption adoption (570).
- **Conventional alternative:** shared server database / SaaS accounts.
- **Peripheral**, but it evidences the single-file, no-backend architecture.

8. Readiness scoring axis (built, removed — alternative embodiment)

- **Claim-like statement:** Scoring each parcel on a set of boolean readiness dimensions (funding, policy/zoning, stakeholder, environmental/title, timeline) and displaying a threshold-colored composite alongside the quantitative projections.
- **Technical steps:** `git show 913a55a:McNichols_Kresge_DEMO.html` lines 218-224 (READINESS), 341 (rdScore), rdColor thresholds 70%/40%; removed in commit 336bbf3.
- **Peripheral / historical**; include for breadth as an alternative decision axis.

9. Intended embodiments (not built — described in TECHNICAL-DISCLOSURE.md §12)

Per-parcel post-investment value (see verdict A); development-mode cost switching on floor area of record; AMI-band-parameterized housing pillar; pro forma gap + funding-pipeline drawdown; three-ring environmental screening from uses of record; before/after concept massing and street elevation from the same assignment state. Each is architecture-specific in §12 and can be claimed as described embodiments.

Clearly conventional — do not claim

- Leaflet map with tile layers, polygons, tooltips, markers (629-758).
- Chart.js bar/doughnut charts (535-538); html2canvas PNG capture (766-776); print-to-PDF via `window.print()` (777).
- localStorage persistence per se (289-300) and JSON file download/upload per se (555-585) — the SCHEMAS and version gating may matter, the browser APIs do not.
- Guided tour overlays (801-830); filter chips; color legends; single-page tabbed HTML app; CSS theming; emoji markers.
- Displaying assessor attributes (owner, zoning, value) on a parcel popup — every county GIS viewer does this.
- Percent-of-AMI affordability arithmetic (income × band%, × 30% ÷ 12) — HUD standard practice.
- Centroid computation, bbox queries, GeoJSON parsing.

Prior art search terms (most → least specific)

1. "parcel-level investment scenario" tool corridor revitalization
2. "cost per outcome" parcel assignment "investment pillar"
3. corridor parcel "scenario comparison" "tax base" projection interactive
4. "post-investment" assessed value projection parcel "value factor"
5. land bank parcel disposition scenario modeling software patent
6. "commercial corridor" revitalization decision support system parcels
7. parcel assignment use type "cost per square foot" outcome projection
8. fiscal impact "new property tax" scenario tool parcel-level interactive
9. mixed-use portfolio rollup parcel projections heterogeneous metrics
10. "scenario planning" GIS parcel "side by side" comparison neighborhood investment
11. TIF district projection interactive parcel tool
12. brownfield corridor reuse planning software parcel scoring
13. site readiness scoring checklist parcel development tool
14. "frontage" proportional corridor diagram visualization parcels schematic
15. offline single-file HTML GIS decision tool localStorage
16. building footprint merge municipal OpenStreetMap dedup centroid
17. scenario JSON export import planning tool version
18. community development financial modeling parcel "area median income"
19. US patent classification G06Q 50/16 (real estate) + scenario simulation
20. US patent classification G06Q 10/0637 (strategic planning) parcel

22. CityEngine / TestFit / Envision Tomorrow scenario "return on investment" parcel
23. CoStar / Regrid / Landgrid parcel analytics patent
24. land use suitability analysis weighted overlay interactive
25. "what-if" land use scenario tax revenue municipal software
26. real estate underwriting automation parcel batch pro forma
27. capital allocation dashboard place-based philanthropy
28. participatory planning tool scenario voting parcels
29. GIS "decision support system" corridor commercial district
30. property value uplift estimation redevelopment model

Suggested searcher note: the closest commercial genre is scenario-planning platforms (UrbanFootprint, Envision Tomorrow, CommunityViz, TestFit) and parcel-data platforms (Regrid, CoStar). The distinguishing axis to probe is (a) parcel-of-record granularity + (b) user-editable coefficient engine + (c) live multi-scenario re-pricing + (d) tax-base output in a single serverless surface, versus their zone/grid-level or fixed-model approaches.

7. License Risk

Prepared 2026-08-03. Attorney work product. Scope: every third-party dependency of the Investment Map files (`McNichols_Kresge_DEMO.html` , `McNichols_Kresge_DEMOv2.html` , `McNichols_Kresge_DEMO_OFFLINE.html`). The tool has NO package manager, NO package.json, NO vendored node_modules — its complete dependency surface is three JavaScript libraries, two font families, and the runtime data/tile services covered in DATA-INVENTORY.md.

1. Dependency table

DEPENDENCY	VERSION	PURPOSE	HOW INCLUDED	LICENSE	EVIDENCE	COPYLEFT?	COMMERCIAL RESTRICTION?
Leaflet	1.9.4	Interactive map, polygons, tiles, markers	CDN `unpkg.com` (v2 lines 7, 222); source inlined in OFFLINE build	BSD-2-Clause	In-repo header in OFFLINE build: "Leaflet 1.9.4, a JS library for interactive maps. https://leafletjs.com" with copyright lines (Volodymyr Agafonkin / CloudMade). BSD-2 identification from the library's standard licensing — UNVERIFIED in repo beyond the header (the full BSD text travels in Leaflet's distribution comment)	No	No — permissive; requires copyright-notice retention
Chart.js	4.4.1	Dashboard bar/doughnut charts	CDN `cdnjs.cloudflare.com` (v2 line 223); inlined in OFFLINE	MIT	In-repo header in OFFLINE build: "Chart.js v4.4.1 … Released under the MIT License". Bundled `@kurkle/color` also MIT (header link github.com/kurkle/color)	No	No
html2canvas	1.4.1	DOM-to-PNG capture for PNG export and PDF report image	CDN `cdnjs.cloudflare.com` (v2 line 224); inlined in OFFLINE	MIT	In-repo header in OFFLINE build: "html2canvas 1.4.1 https://html2canvas.hertzen.com Copyright (c) 2022 Niklas von Hertzen … Released under the MIT License"	No	No
tslib (bundled inside html2canvas)	n/a	TypeScript runtime helpers	Transitively inlined in OFFLINE	0BSD	In-repo header: "Copyright (c) Microsoft Corporation." (tslib's standard 0BSD grant; exact SPDX UNVERIFIED in repo — the header text present is the permission notice)	No	No
Fraunces (font)	via Google Fonts	Display/heading typeface	CSS link, live fetch (v2 line 10; OFFLINE still fetches fonts remotely)	Expected SIL OFL 1.1 — UNVERIFIED (no license in repo)	RESEARCH REQUIRED	OFL is copyleft-for-fonts only (derivative FONTS must be OFL); does not affect the app	No for embedding/use
Inter (font)	via Google Fonts	Body/UI typeface	Same	Expected SIL OFL 1.1 — UNVERIFIED	RESEARCH REQUIRED	Same as above	No for embedding/use

Runtime services (not code dependencies; full detail in DATA-INVENTORY.md): OSM tiles + Overpass data (ODbL — see §3), Esri World_Imagery tiles (terms RESEARCH REQUIRED), Detroit ArcGIS open data (RESEARCH REQUIRED), Google Fonts CDN, unpkg/cdnjs CDN terms (RESEARCH REQUIRED), HUD AMI figure (RESEARCH REQUIRED, expected public domain).

The other projects in this repository (`docs/` , `hlb-7811-harrisburg-rfp/`) are documentation-only (Markdown) with no code dependencies, and are out of scope for this disclosure.

2. Copyleft flags

- **No GPL, AGPL, LGPL, MPL, or other code copyleft anywhere in the dependency surface.** All three JS libraries are permissive (BSD-2/MIT).
- **SIL OFL (fonts):** copyleft applies only to derivative font software, not to documents or apps using the fonts. No obligation triggered by this tool's usage pattern. (Verify the Google Fonts serving terms — RESEARCH REQUIRED.)

renders OSM tiles with attribution (v2:651), and (3) fetches OSM building footprints via Overpass, MERGES them with City of Detroit footprints after centroid dedup (v2:701-713), and caches the merged set in the user's browser (v2:662). ODbL requires attribution and, for publicly "used" derivative databases, share-alike of the derivative database. Counsel question: is the merged, cached footprint set a "derivative database" that is "publicly used" when the tool is sold/hosted? Mitigations if needed are cheap (attribute OSM in the footprint note, or drop the OSM supplement and use city data only). This does not encumber the tool's CODE or its inventive method — it touches only the footprint overlay data.

3. Does anything block selling this as hosted software?

From the code dependencies: no. BSD-2 and MIT expressly permit commercial use, resale, and hosting; the only obligation is preserving copyright/license notices — already satisfied in the OFFLINE build (headers retained) and satisfied for the CDN build by not redistributing the libraries at all.

From the data/services layer: three items need resolution before commercial hosting, none of which look structural:

1. **Esri World_Imagery tiles** (the default basemap) — most likely to require a paid ArcGIS plan or richer attribution for commercial apps. Fallback exists in-product (OSM basemap toggle, v2:724). RESEARCH REQUIRED.
2. **OSM tile usage policy + ODbL share-alike** for the merged footprint cache (§2). Fixable with attribution and/or source selection. RESEARCH REQUIRED.
3. **City of Detroit assessor/open-data terms** for redistributing parcel records (including owner names) inside a commercial deliverable. The tool already carries the provenance stamp (v2:228). RESEARCH REQUIRED.

Plain statement: **nothing in the repository imposes a copyleft or non-commercial restriction on the inventive software itself; the open questions are all data-service terms (Esri, OSM/ODbL, City of Detroit), each of which has an in-architecture fallback or a trivial attribution fix.**

UNVERIFIED items: exact SPDX texts for Leaflet/tslib beyond the headers quoted (the repo carries headers, not full LICENSE files, for the inlined libraries); all RESEARCH REQUIRED rows above. Per instructions, no online license research was performed.

Prepared from a read-only review of the codebase; no source files were modified. Flowcharts are provided as Mermaid diagrams (rendered when online; shown as source text otherwise) — the underlying logic is fully described in the Technical Disclosure section regardless. This document is documentation support for counsel, not legal advice.